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| **Fiscal Year:** | FY 2025 | **Task Last Updated:** | FY 07/15/2026 |
| **PI Name:** | Burke, Shawn Ph.D. | | |
| **Project Title:** | Facilitating the Synergistic Side of Cultural Differences in LDSE: Identification of Challenges and Development of Cultural Training | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Behavior and performance | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H) | | |
| **Human Research Program Risks:** | (1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture<br>(2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 32826-3281 | **Congressional District:** | 7 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2014-15 HERO NNJ14ZSA001N-MIXEDTOPICS. Appendix E: Behavioral Health & Human Health Countermeasures Topics |
| **Start Date:** | 08/22/2016 | **End Date:** | 07/31/2025 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 6 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Whitmire, Alexandra | **Contact Phone:** | |
| **Contact Email:** | alexandra.m.whitmire@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 7/31/2025 per L.Dang/HFBP (Ed., 4/14/25)<br>NOTE: End date changed to 5/31/2025 per D. Kulkarni/HFBP (Ed., 7/12/24)<br>NOTE: End date changed to 8/21/2024 per NSSC information (Ed., 8/23/23)<br>NOTE: End date changed to 8/21/2023 per NSSC information (Ed., 7/22/22)<br>NOTE: End date changed to 8/21/2022 per NSSC information (Ed., 8/24/21)<br>NOTE: End date changed to 8/21/2021 per NSSC information (Ed., 7/27/20) | | |
| **Key Personnel Changes/Previous PI:** | July 2025 report: No key personnel changes | | |

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| **COI Name (Institution):** | Salas, Eduardo Ph.D. ( Rice University ) |
| **Grant/Contract No.:** | NNX16AP96G |
| **Performance Goal No.:** | |
| **Performance Goal Text:** | |
| **Task Description:** | Culture has been a factor for space exploration since the mid 1970s when the U.S. first partnered with Russia. Work has indicated that culturally-based differences in values, beliefs, and preferences for cognition and action can have an impact on interdependent action. Cultural differences can pose challenges for a number of transition, action, and interpersonal processes and emergent states in teams. However, almost all of this work has been conducted outside the context of spaceflight. The little work that has been conducted within spaceflight has begun to suggest that the impact of culture on teams may be different in long-duration space exploration (LDSE). This drives the question as to whether cultural differences within spaceflight crews has the same impact on team performance as that reported within the wider literature on cross-cultural teams and what can be done to create synergy. We leverage prior work on cultural differences and teams in mission critical environments and archival data on team interaction in ICE (isolated, confined environments) and pair this with experimentation in ground analog studies to better understand the critical challenges created by cultural differences in LDSE and use this information to create and test cultural training tools that can be used pre- and in-flight. Specifically, we take a multi-pronged approach to answer the following questions: (1) what are the critical issues surround culture and team performance in LDSE, (2) from a compositional standpoint is there an 'ideal' team profile in terms of cultural orientations to facilitate team performance and adaptability in LDSE, (3) what types of cultural training can be developed for use pre-flight and in-flight, and (4) what tools can be developed to facilitate practitioners in maximizing the potential synergy in culturally different teams operating in LDSE (e.g., pillars, guidelines, practices, tips). |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Within this project, we seek to answer the following questions: (1) what are the critical issues surrounding culture and team performance in long-duration spaceflight, (2) is there a team cultural profile that facilitates team performance and adaptation in long-duration spaceflight, and (3) what tools can be developed to facilitate synergy/mitigate decrements in teamwork and team performance within culturally different teams operating in the context of spaceflight.<br>The proposed project will impact numerous areas. First, this research will address current theoretical gaps surrounding the critical challenge of cultural differences in long-duration spaceflight (both on the ground and in orbit). This information is then used to build a series of complementary, modular training tools to facilitate the crew's ability to turn cultural differences into a synergistic force. The existent literature on cross-cultural training does little to focus on cultural differences in crews operating in isolated, confined environments. Additionally, the focus of most cross-cultural training programs is to prepare expatriates to go abroad; therefore, teaching cross-cultural competence with regard to teamwork skills is rarely seen. In building our training we seek to validate the effectiveness of such programs within analog environments using sound scientific methods and training evaluation techniques. Finally, as an important part of training development is the assessment of the skills gained during training we will also deliver a set of assessment tools that can be used to diagnose cross-cultural competence levels with regard to team interaction. |
| **Task Progress:** | The research focus within the grant was to begin to examine the impact of cultural difference on team interaction (processes, emergent states) and performance in long-duration space exploration (LDSE) and then use this understanding to develop and test a training countermeasure. In doing so, we sought to answer the following questions: (1) what are the critical issues surrounding culture and team performance in long-duration spaceflight, (2) from a compositional standpoint is there an 'ideal' team profile in terms of cultural orientations to facilitate team performance and adaptability in LDSE, (3) what types of cultural training can be developed, (4) what tools can be developed to facilitate practitioners in maximizing the potential synergy in culturally different teams operating in LDSE (e.g., pillars, guidelines, practices, tips). In answering these questions, we sought to provide a series of scientifically grounded guidelines, tools, and cultural training which serve to mitigate challenges in culturally different crews working in LDSE and allow crews to capitalize on the potential synergy present.<br>We took a multi-pronged approach to explore these questions, whereby data was collected from several different sources (i.e., analog environments, interviews, archived information – NASA's Oral History Project). Analog data collection involved multicultural, mixed crews ranging from 4-6 members (analog dependent) who engaged in mission-relevant tasks for periods of time ranging from 45 days to 8 months (mission dependent). Moreover, one-on-one interviews were conducted with crewmembers upon the conclusion of the mission. To examine the collected data, both qualitative and quantitative analytic approaches were applied.<br><br>The Nature of the Crews<br><br>To contextualize our findings, it is important to acknowledge that analog crews were low to moderate in differences, with respect to the cultural values examined (i.e., collectivism, power distance, uncertainty avoidance, masculinity, context, long-term orientation; see Hofstede, 2001; Hall & Hall, 1990; Holtgraves, 1997). The degree of difference with regard to cultural dimensions was computed using the work of Harrison and Klein (2007).<br><br>Type of Cultural Differences Uncovered<br><br>Based on data extracted from both interviews and journals findings, there were a number of salient differences which emerged within the crews, including those related to: (1) beliefs, customs, and traditions; (2) language; (3) interpersonal orientation and team dynamics; and (4) interpersonal relations, (5) general cultural background, and (6) organizational culture of national agencies. These differences are in addition to the cultural profiles which were built for the crew, indicating that crewmembers differed in the targeted cultural values – with some cultural values witnessing greater variability than others.<br><br>Role of Culture on Team Dynamics<br><br>Data indicated that cultural values and beliefs were shown to have an impact on the manner in which the crew interacted – impacting both team process (e.g., leadership, teamwork, conflict management) and emergent states (e.g., trust, team |

identity). For example, the degree to which crewmembers valued a team structure, such that status served to govern team interactions (i.e., cultural value of high power distance), was related to lower levels of trust and teamwork, as well as lower levels of team identity. Additionally, valuing this type of interaction structure was related to members perceiving more faultlines within the crew, lower levels of team identity, trust, and teamwork.

With regard to cultural differences, there were several team processes that were salient to crewmembers, including but not limited to: communication, leadership, teamwork, and monitoring/backup. Other team processes were also salient with respect to cultural differences, but not to the extent as the ones named above. While not a team process, language was also frequently mentioned as being salient. The data suggested that it was not only team processes that were salient in relation to cultural differences, but emergent states also.

Data also indicated the role of context. Some of the results that emerged include: (1) isolation may sharpen one's sense of ethnic background; (2) family metaphors/dynamics may be more frequent in teams operating in isolated, confined environments – even when crews are multicultural; (3) the specific impact of particular cultural values on team dynamics varies over time and is context dependent. For example, results indicated that while cultural differences appeared through the mission, they were most often mentioned/salient in the first half of the mission as compared to the second half.

The Building of Synergy Within Multicultural Crews

Finally, the project revealed a number of mechanisms that crews utilize to transmit cultural knowledge and build cohesion. Utilizing the information gained throughout the project in terms of the impact of cultural difference and associated values on team interaction, as well as identified leverage points, a training program was developed to facilitate the building of shared mental models within the crew. The cultural training program had the following learning objectives: 1) identify the importance of paying attention to this compositional feature within the crew; (2) understand how individuals may differ in their values, beliefs, and attitudes based on their national culture; (3) understand how cultural values and corresponding differences may manifest to impact crew interaction; (4) recognize the value of team metaphors; and (5) identify and apply techniques to maximize the positive effects of cultural difference. Training evaluation results indicated that, overall, the crew found utility in the training, liked the training and would recommend it, found it logically organized, and felt they learned new knowledge and techniques.

| **Bibliography Type:** | Description: (Last Updated: 07/15/2026) |
|---|---|
| **Articles in Peer-reviewed Journals** | Marquez JJ, Landon LB, Salas E. "The next giant leap for space human factors: The opportunities." Hum Factors. 2023 May 28;187208231174955. https://doi.org/10.1177/00187208231174955 ; PMID: 37246369 , May-2023 |
| **Conference Materials (Downloadable)** | Szabo, K., Kaosaar, A., Burke, C. S. "Orbiting cultures: Examining cultural diversity in analog crews over time" Abstracts. 2025 NASA Human Research Program Investigators Workshop, Galveston, Texas, January 28-31 , Jan-2025 |
| **Conference Materials (Downloadable)** | Burke, C. S. "How the social and cognitive sciences are exploring collaboration in and for long duration space travel." Space Translational Advances & Research [STAR Nona]. Space: The final frontier of medicine. Lake Nona Cancer Center, Orlando, FL. , Apr-2026 |
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